

Sketching Graphs of Quadratic Functions

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A **quadratic function** has the general form

$$y = ax^2 + bx + c, \quad a \neq 0$$

Its graph is called a **parabola**.

Opening Direction

- ▶ If $a > 0$, the parabola **opens upward**.
- ▶ If $a < 0$, the parabola **opens downward**.

The Vertex

The **vertex** is the highest or lowest point of the parabola.

- ▶ It is the **lowest** point when the parabola opens upward.
- ▶ It is the **highest** point when the parabola opens downward.

Axis of Symmetry

The **axis of symmetry** is the vertical line passing through the vertex.

$$x = -\frac{b}{2a}$$

y -Intercept

The **y -intercept** is found by letting $x = 0$.

$$y = a(0)^2 + b(0) + c = c$$

x -Intercepts

The x -**intercepts** (if they exist) are the zeros of the quadratic function.

They are found by letting $y = 0$ and solving for x .

Sketching a Parabola

A graph can be sketched by identifying:

- ▶ the **vertex**,
- ▶ the **axis of symmetry**,
- ▶ the **intercepts**, and
- ▶ a few additional **symmetric points**.

Symmetry of the Graph

The graph is always **symmetric** about its axis of symmetry.

Points on one side of the axis have matching points, the same distance away, on the other side.

A Neat Sketch

A neat sketch should include:

- ▶ labeled axes,
- ▶ the intercepts,
- ▶ the vertex, and
- ▶ the direction in which the parabola opens.

Examples

Example 1

Analyze $y = x^2 - 2x - 3$.

Opens: upward ($a = 1 > 0$)

Axis of symmetry: $x = -\frac{-2}{2(1)} = 1$

y -intercept: $(0, -3)$

x -intercepts: $(x-3)(x+1) = 0 \rightarrow (3, 0), (-1, 0)$

Vertex: $(1, -4)$

Example 2

Analyze $y = -x^2 + 4x - 3$.

Opens: downward ($a = -1 < 0$)

Axis of symmetry: $x = -\frac{4}{2(-1)} = 2$

y -intercept: $(0, -3)$

x -intercepts: $-(x-1)(x-3) = 0 \rightarrow (1, 0), (3, 0)$

Vertex: $(2, 1)$

Practice Exercises

Practice 1

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 4x + 3$$

Practice 2

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 2x - 8$$

Practice 3

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 6x - 5$$

Practice 4

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 2x^2 - 8x + 6$$

Practice 5

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -2x^2 - 4x + 6$$

Practice 6

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 6x + 5$$

Practice 7

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 10x + 21$$

Practice 8

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 4x$$

Practice 9

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 3x^2 - 12x + 9$$

Practice 10

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 - 2x + 8$$

Activities

Activity 1

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 6x + 5$$

Activity 2

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 4x + 3$$

Activity 3

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 8x - 12$$

Activity 4

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 2x^2 - 4x - 6$$

Activity 5

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 - 8x + 12$$

Activity 6

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = x^2 + 8x + 15$$

Activity 7

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 + 2x + 3$$

Activity 8

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = 3x^2 + 6x - 9$$

Activity 9

Sketch the graph. Label the vertex, axis of symmetry, intercepts, and direction of opening.

$$y = -x^2 - 6x + 7$$

Thank you!